MATH 111 Calculus 1

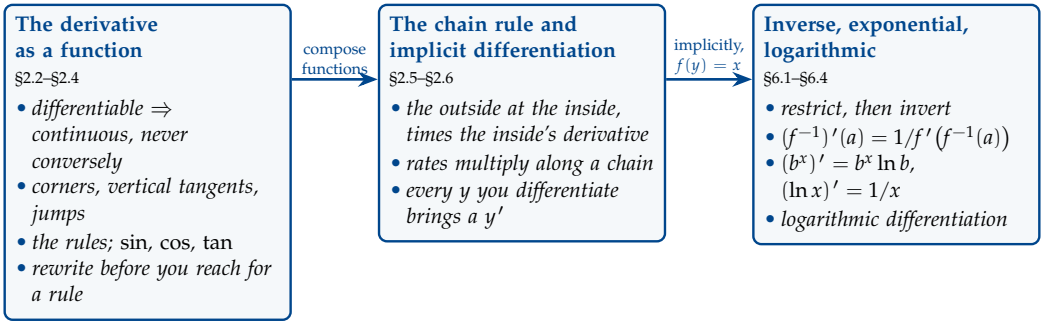
Revision

The derivative as a function, the rules, inverse, exponential and logarithmic functions — in questions

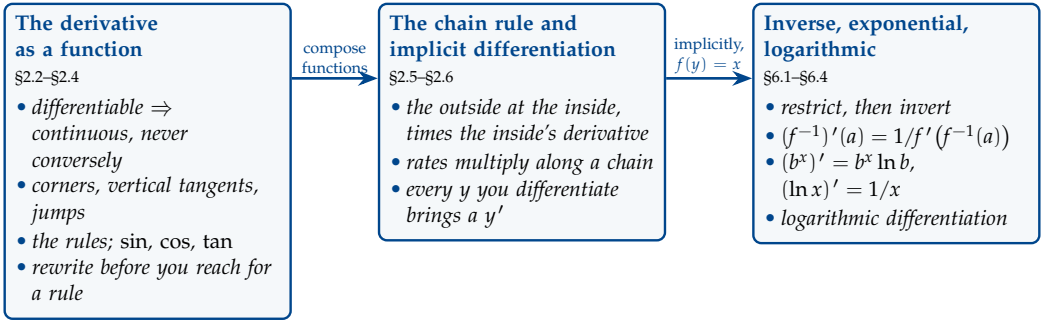
Week 7 · Lecture 18 · §2.2–§2.6, §6.1–§6.4

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

The map — §2.2–§2.6 and §6.1–§6.4 on one page



The map — §2.2–§2.6 and §6.1–§6.4 on one page



Today

Every question today is a rule you know, used in the right order: name the outside, find the point — and where two formulas meet, go back to the definition.

STAND UP

"If f is one-to-one and differentiable on $\mathbb{R}$, then f^{-1} is differentiable at every point of its domain."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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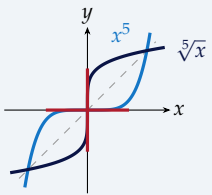
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Verdict

FALSE. $f(x) = x^5$ is increasing, so one-to-one, and differentiable everywhere, but $f'(0) = 0$. Its inverse $\sqrt[5]{x}$ at 0:

$$\lim_{h\to 0} \frac{\sqrt[5]{h} - 0}{h} = \lim_{h\to 0} \frac{1}{\left(\sqrt[5]{h}\right)^4} = \infty$$ — a vertical tangent, no derivative. The formula $(f^{-1})'(a) = 1/f'\left(f^{-1}(a)\right)$ needs $f'\left(f^{-1}(a)\right) \neq 0$: where the graph of f is flat, its mirror image in $y = x$ — the graph of f^{-1} — is vertical.



red: the tangents at 0

Question 53

Let $f(x) = \begin{cases} e^x - 1, & x \le 0, \\ \sin x, & x > 0. \end{cases}$ At $x = 0$ the function f is

A. not continuous: $e^x - 1$ and $\sin x$ are different functions.

B. continuous, and differentiable with $f'(0) = 1$.

C. continuous, but not differentiable: two different formulas meet at 0.

D. continuous, and differentiable with $f'(0) = 0$, because $f(0) = 0$.

In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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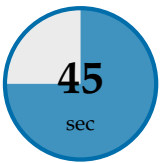
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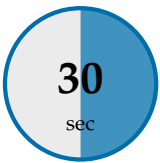
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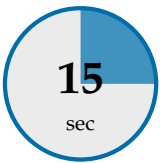
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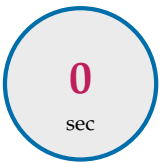
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Hands down. Thank you.

A B C D

Answer

B. Continuous: both one-sided limits are $0 = f(0)$. The slopes, one side at a time: $\lim_{h\to 0^-} \frac{e^h - 1}{h} = 1$ (the number e) and $\lim_{h\to 0^+} \frac{\sin h}{h} = 1$: equal, so $f'(0) = 1$ (frame 7).

C: a corner needs two *different* one-sided slopes.

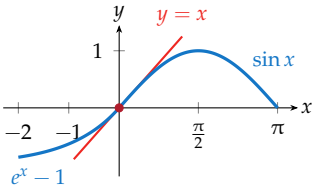
A compares formulas, not values.

D differentiates the number $f(0)$.

Question 53 — the derivative at the seam, written out

$f(x) = e^x - 1$ for $x \le 0$, $\quad f(x) = \sin x$ for $x > 0$.

Continuity at 0, three conditions: $f(0) = e^0 - 1 = 0$;
$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-}(e^x - 1) = 0$ and
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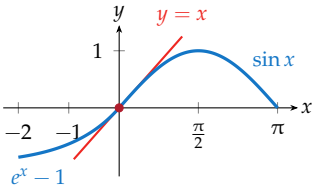
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PROOF that $f'(0) = 1$ — one side at a time.



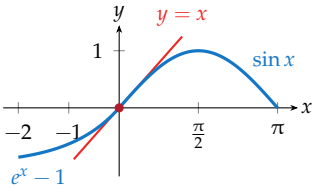
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For $h < 0$, $f(0+h) = e^h - 1$: $\quad \lim_{h\to 0^-} \dfrac{f(0+h) - f(0)}{h} = \lim_{h\to 0^-} \dfrac{(e^h - 1) - 0}{h} = 1;$

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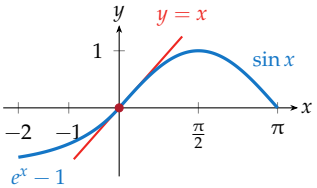
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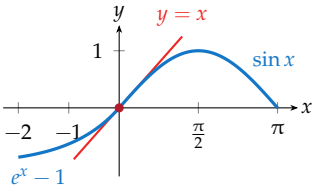
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Both one-sided limits equal 1, so the limit exists: $f'(0) = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = 1.$ ■

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Used: $\lim_{h\to 0} \frac{e^h - 1}{h} = 1$ (the definition of e) and $\lim_{h\to 0} \frac{\sin h}{h} = 1$. For $x \neq 0$ the formulas give $f'(x) = e^x$ and $\cos x$, both 1 at 0 — consistent; but at the seam the definition decides.

Question 54

Suppose $f(-1) = 4$, $f'(-1) = 1$, $g(-1) = 2$, $g'(-1) = -5$. What is $\left(\frac{f}{g}\right)'(-1)$?

A. $-\frac{11}{2}$

B. $-\frac{1}{5}$

C. 11

D. $\frac{11}{2}$

In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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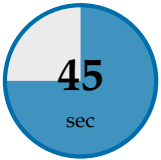
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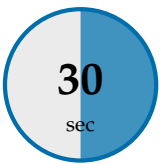
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Hands down. Thank you.

A B C D

ANSWER

Answer

D. $\left(\frac{f}{g}\right)'(-1) = \frac{g(-1)f'(-1) - f(-1)g'(-1)}{g(-1)^2} = \frac{2 \cdot 1 - 4 \cdot (-5)}{2^2} = \frac{22}{4} = \frac{11}{2}.$

A reverses the numerator, $fg' - gf'$: the bottom times the derivative of the top, minus the top times the derivative of the bottom.

B divides the derivatives, f'/g' — no rule says that.

C forgets to square the denominator.

STAND UP

"At $x = e$, the graphs of $y = e^x$ and $y = x^e$ have the same tangent line."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP

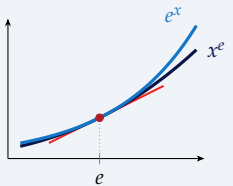
"At $x = e$, the graphs of $y = e^x$ and $y = x^e$ have the same tangent line."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

Verdict

TRUE. Both pass through (e, e^e). Slopes: $(e^x)' = e^x$, which is e^e at $x = e$; $(x^e)' = e\,x^{e-1}$ — the power rule: the exponent is constant — which is $e \cdot e^{e-1} = e^e$ at $x = e$. Same point, same slope: the curves touch there.



they touch at (e, e^e)

Question 55

Error hunt. Find y' for $y = (2 + \sin x)^x$. Which line is the *first* wrong one?

Line 1: $\ln y = x \ln(2 + \sin x)$.

Line 2: $\dfrac{y'}{y} = \ln(2 + \sin x) + x \cdot \dfrac{\cos x}{2 + \sin x}$.

Line 3: So $y' = \ln(2 + \sin x) + \dfrac{x \cos x}{2 + \sin x}$.

Line 4: At $x = \dfrac{\pi}{2}$: $y' = \ln 3$.

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4

In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

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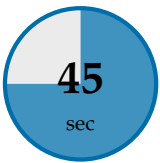
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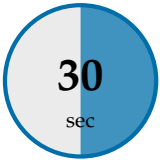
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A B C D

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A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4

Hands down. Thank you.

A B C D

Answer

C. Line 3 forgets to multiply by y: $y' = (2+\sin x)^x\left(\ln(2+\sin x) + \frac{x\cos x}{2+\sin x}\right)$.

A, **B**: lines 1 and 2 are right ($2+\sin x \geq 1 > 0$; the product rule, and the chain rule on $\ln(2+\sin x)$).

D: line 4 follows correctly from the wrong line 3; the true value at $\frac{\pi}{2}$ is $3^{\pi/2}\ln 3 \approx 6.17$, not $\ln 3$.

Question 56

$f(x) = x^3 + 10x - 27$ is one-to-one. What is $\left(f^{-1}\right)'(1)$?

A. $\dfrac{1}{22}$

B. 22

C. $\dfrac{1}{13}$

D. It cannot be found: there is no formula for f^{-1}.

In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

Question 56

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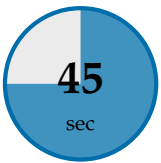
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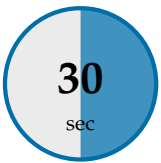
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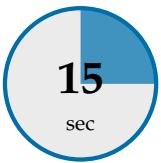
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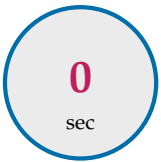
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A B C D

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Hands down. Thank you.

A B C D

Answer

A. First the point: $f(2) = 8 + 20 - 27 = 1$, so $f^{-1}(1) = 2$. Then $(f^{-1})'(1) = 1/f'\big(f^{-1}(1)\big) = 1/f'(2) = \frac{1}{22}$, since $f'(x) = 3x^2 + 10$.

B forgets the reciprocal.

C feeds f' the number 1 itself ($f'(1) = 13$): f' wants $f^{-1}(1) = 2$.

D: no formula for f^{-1} is needed — only the point, and the slope there.

Question 57

Which function f satisfies $f'(x) = -2f(x)$ for every x?

A. $f(x) = -2e^x$

B. $f(x) = e^{-2x}$

C. $f(x) = 2^{-x}$

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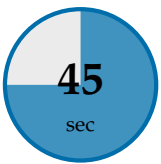
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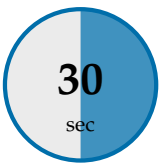
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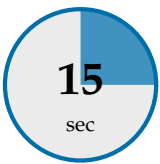
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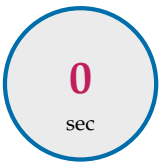
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Hands down. Thank you.

A B C D

Answer

B. $(e^{-2x})' = e^{-2x} \cdot (-2) = -2e^{-2x}$: the chain rule, inside $-2x$.

A: $(-2e^{x})' = -2e^{x} = f$, not $-2f$: the -2 in front is part of f, and stays there.

C: $(2^{-x})' = -(\ln 2)\, 2^{-x}$, not $-2 \cdot 2^{-x}$: $\ln 2$ comes down, not the base.

D: $(e^{-2x} + 1)' = -2e^{-2x}$, but $-2f(x) = -2e^{-2x} - 2$: the constant 1 has derivative 0, yet it is part of f on the right-hand side — check the equation, not only the derivative.

STAND UP

"For every $x \neq 0$: $\dfrac{d}{dx} \ln |x| = \dfrac{1}{|x|}$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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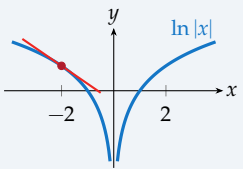
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Verdict

FALSE. For $x < 0$: $\ln|x| = \ln(-x)$, and the chain rule gives $\dfrac{1}{-x}\cdot(-1) = \dfrac{1}{x}$ — negative, as it must be: as x increases towards 0 from the left, $|x|$ decreases, so $\ln|x|$ decreases. For every $x \neq 0$: $(\ln|x|)' = \dfrac{1}{x}$. At $x = -2$: slope $-\frac{1}{2}$, not $\frac{1}{2}$.



at $x = -2$ the slope is $-\frac{1}{2}$

Question 58

The point $(2, -2)$ lies on the curve $y^2 = x^3 + xy$. The slope of the tangent there is

A. -7

B. -2

C. -5

D. $-\frac{5}{3}$

In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

The point $(2, -2)$ lies on the curve $y^2 = x^3 + xy$. The slope of the tangent there is

A. -7

B. -2

C. -5

D. $-\frac{5}{3}$



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THINK & TALK · 1 minute

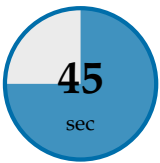
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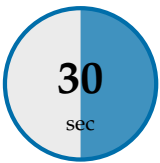
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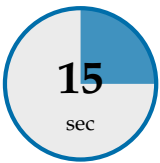
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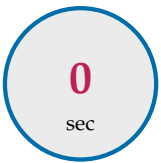
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A B C D

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Hands down. Thank you.

A B C D

Answer

D. Differentiate as it stands — every y you differentiate brings a y', and xy is a product: $2y\,y' = 3x^2 + y + x\,y'$. So $y'(2y - x) = 3x^2 + y$ and

$$y' = \frac{3x^2 + y}{2y - x} = \frac{12 - 2}{-4 - 2} = -\frac{5}{3}.$$

A drops the y' after y^2: $2y = 3x^2 + y + xy'$ gives -7.

B differentiates xy as $x\,y'$: $y'(2y - x) = 3x^2$, -2.

C moves xy' across with the wrong sign: $y'(2y + x) = 3x^2 + y$, -5.

(Check first: the point is on the curve, $(-2)^2 = 4 = 8 - 4$.)

Question 59

True or false ?

If f and g are differentiable and $(fg)'(a) = 0$,
then $f'(a) = 0$ or $g'(a) = 0$.

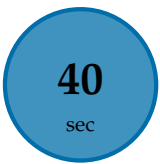
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Decide your own answer first — true or false — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 40 seconds

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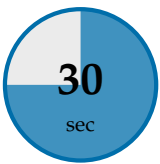
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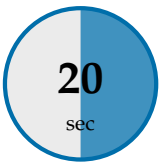
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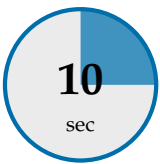
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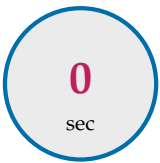
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TRUE FALSE

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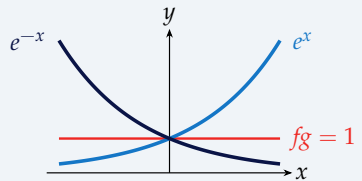
TRUE FALSE

ANSWER

Answer

False. $f(x) = e^x$ and $g(x) = e^{-x}$:
$f(x)\,g(x) = e^0 = 1$ for every x, so $(fg)'(a) = 0$ at every a — but $f'(a) = e^a \neq 0$ and $g'(a) = -e^{-a} \neq 0$.

The product rule says why:
$(fg)'(a) = f'(a)\,g(a) + f(a)\,g'(a)$ — two terms, which can cancel.



e^x rises, e^{-x} falls; the product is flat

Question 60

Let $g(x) = f(\sin x)$, where f is differentiable and $f'\left(\frac{1}{2}\right) = 4$. What is $g'\left(\frac{\pi}{6}\right)$?

A. 4

B. 2

C. $2\sqrt{3}$

D. Not enough information: $f'\left(\frac{\pi}{6}\right)$ is needed.

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THINK & TALK · 1 minute

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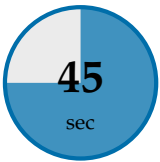
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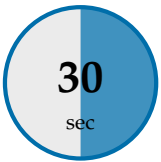
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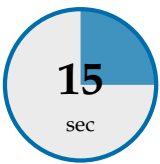
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Hands down. Thank you.



Answer

C. Outside f, inside $\sin x$: $g'(x) = f'(\sin x) \cdot \cos x$, so $g'\left(\frac{\pi}{6}\right) = f'\left(\frac{1}{2}\right) \cdot \cos\frac{\pi}{6} = 4 \cdot \frac{\sqrt{3}}{2} = 2\sqrt{3}$.

A forgets the inside's derivative.

B multiplies by $\sin\frac{\pi}{6} = \frac{1}{2}$ — the inside itself instead of its derivative.

D feeds the outside's derivative x instead of the inside $\sin x$: f' is needed at $\sin\frac{\pi}{6} = \frac{1}{2}$, which is given.

By heart (1) — the rules

The rules

$(c)' = 0, \quad (x^n)' = nx^{n-1}$ (every real n),

$(cf)' = cf', \quad (f \pm g)' = f' \pm g',$

$(fg)' = fg' + gf', \quad \left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2};$

$(\sin x)' = \cos x, \quad (\cos x)' = -\sin x,$

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Implicit, inverse, exponential, logarithmic

Differentiate both sides — every y you differentiate brings a y' — then solve for y'.

$(f^{-1})'(a) = \dfrac{1}{f'\big(f^{-1}(a)\big)}$: find the point first.

$(e^x)' = e^x, \quad (b^x)' = b^x \ln b,$

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The habits

Rewrite before you reach for a rule. Say "outside" before you differentiate a composition. Check that the point is on the curve. Differentiable implies continuous — never conversely.

By heart (2) — logarithms and powers

Logarithmic differentiation, three steps

1. Take ln of both sides and use the laws. **2.** Differentiate implicitly. **3.** Multiply by y.

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Which rule? Is the base constant, or the exponent?

b^n	constant, constant	0
$\big[f(x)\big]^n$	variable, constant	$n\big[f(x)\big]^{n-1} f'(x)$ (the power rule, with a chain)
$b^{g(x)}$	constant, variable	$b^{g(x)} \ln b \cdot g'(x)$
$\big[f(x)\big]^{g(x)}$	variable, variable	logarithmic differentiation

By heart (2) — logarithms and powers

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$\big[f(x)\big]^{g(x)}$	variable, variable	logarithmic differentiation

Logarithms

For $x, y > 0$: $\ln(xy) = \ln x + \ln y$, $\ln \frac{x}{y} = \ln x - \ln y$, $\ln(x^r) = r \ln x$. $e^{\ln x} = x$ $(x > 0)$, $\ln(e^x) = x$ (every x).

$\ln(x^2) = 2\ln|x|$ for $x \neq 0$. The number e: the base whose exponential has slope 1 at 0.

STAND UP

"If $f'(x) = f(x)$ for every x, then $f(x) = e^x$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP

"If $f'(x) = f(x)$ for every x, then $f(x) = e^x$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

Verdict

FALSE. $f(x) = 3e^x$ — and even $f(x) = 0$.

Why: every such f is Ce^x — $\left(f(x)\,e^{-x}\right)' = f'(x)\,e^{-x} - f(x)\,e^{-x} = 0$, so $f(x)\,e^{-x}$ is constant on $\mathbb{R}$ (a function with derivative 0 on an interval is constant).

Tonight — three things

1. Ten derivatives of your own choosing, one of each kind, without notes: a product; a quotient; a chain of three; implicit; an inverse at a point; $b^{g(x)}$; $\ln|u|$; logarithmic differentiation; a variable base and a variable exponent; a derivative at a seam, from the definition.

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Next:

inverse trigonometric and hyperbolic functions, extreme values, the Mean Value Theorem and the shape of a graph — in questions.

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *the one kind of derivative that I still cannot do is … (a kind of question, not a chapter).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.